Design of a 4-Stage Pipelined Processor

Internship Task 3 – CODTECH

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Project Title: Design and Simulation of a 4-Stage Pipelined Processor using Verilog HDL

Software Used: Xilinx Vivado Language: Verilog HDL

1. Introduction

A pipelined processor improves system performance by executing multiple instructions simultaneously in different stages of execution. Instead of completing one instruction at a time, pipelining divides instruction execution into stages, allowing parallel processing.

This project implements a **4-stage pipelined processor** supporting basic instructions such as **ADD, SUB, and LOAD** using Verilog HDL.

2. Objective

- To design a basic pipelined processor.
- To implement multiple execution stages.
- To understand instruction flow in pipelining.
- To verify processor functionality using simulation waveform.

3. Pipeline Architecture

The processor contains four stages:

Stage 1 — Instruction Fetch (IF)

- Fetches instruction from instruction memory.
- Program Counter (PC) increments.

Stage 2 — Instruction Decode (ID)

- Decodes opcode.
- Transfers instruction to execution stage.

Stage 3 — Execute (EX)

- Performs arithmetic or logical operation.

Stage 4 — Write Back (WB)

- Stores result into register file.

Pipeline registers used:

- IF_ID
- ID_EX
- EX_WB

4. Instruction Format

Bits	Operation
00	ADD
01	SUB
10	LOAD

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5. Design Description

The processor consists of:

- Instruction Memory
- Register File
- Arithmetic Logic Unit (ALU)
- Pipeline Registers
- Program Counter

Each clock cycle moves instructions forward through the pipeline stages.

6. Verilog Implementation

The design is implemented using Verilog HDL in Xilinx Vivado.

Main modules:

- pipeline.v — Processor Design
- pipeline_tb.v — Testbench

The testbench generates clock and reset signals to simulate processor execution.

7. Simulation Setup

Simulation steps:

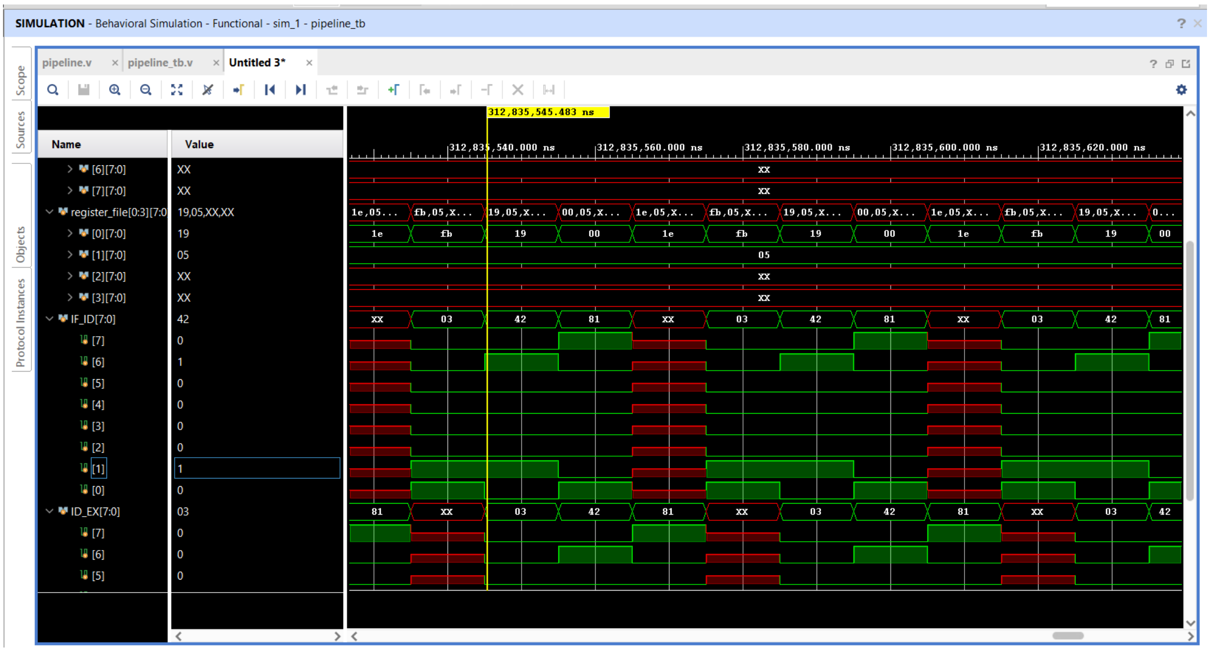
1. Created project in Vivado.
2. Added design source and testbench.
3. Set testbench as Top Module.
4. Ran Behavioral Simulation.

5. Added internal pipeline signals to waveform.

Signals observed:

- clk
- reset
- pc
- IF_ID
- ID_EX
- EX_WB

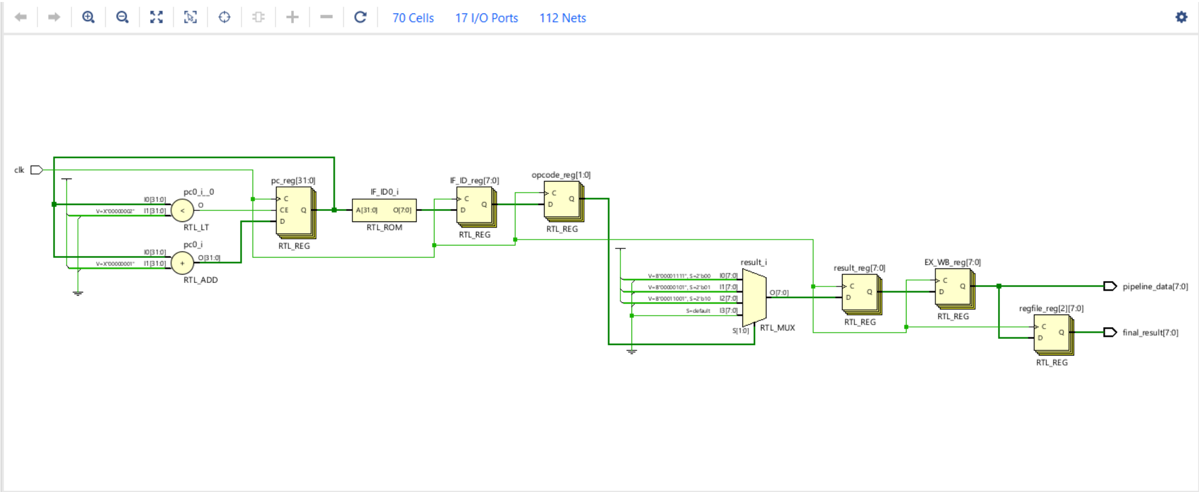
8. Simulation Results



Observation:

- Clock signal toggles continuously.
- Instructions propagate across pipeline stages.
- Multiple instructions execute simultaneously.
- Correct pipeline behavior achieved.

9. RTL Schematic



10. Observations

- Pipeline registers introduce stage delays.
- Throughput improves compared to single-cycle execution.
- Each stage operates concurrently after pipeline fill.

11. Advantages of Pipelining

- Increased processor performance
- Parallel instruction execution
- Efficient hardware utilization
- Reduced instruction execution time

12. Conclusion

The 4-stage pipelined processor was successfully designed and simulated using Verilog HDL. Simulation results verified correct instruction flow through Fetch, Decode, Execute, and Writeback stages. The project demonstrates the fundamental concept of pipelining used in modern processors.

13. References

- Xilinx Vivado Documentation
- Digital Design and Computer Architecture Textbook
- Verilog HDL Reference Guide